Project Proposal: Risk-Adjusted Portfolio Optimization (ML vs. Markowitz)

Team Members: *Alexy Skoutnev, Shaun Mukahanana, Tadiwanashe Chiremba*

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1. Project Description

Abstract

The Markowitz Mean-Variance Optimization (MVO) framework (Markowitz, 1952) provides a foundational approach to portfolio selection but is sensitive to estimation error and assumes linear relationships among asset returns. Deep learning models, particularly Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997), have shown strong performance in modeling nonlinear temporal dependencies in financial markets. This project evaluates whether LSTM-based forecasting can improve risk-adjusted returns—measured using Sharpe and Sortino ratios—and reduce downside risk during high-volatility regimes.

Introduction

Classical MVO (Markowitz, 1952) remains central to portfolio theory, yet its practical limitations motivate the exploration of nonlinear models. Financial markets often exhibit regime-dependent behavior, making deep learning models attractive due to their ability to capture temporal and nonlinear patterns. LSTM networks have demonstrated strong predictive performance in financial time series forecasting (Fischer & Krauss, 2018). This project investigates whether LSTM-based return forecasting can enhance portfolio construction relative to MVO, particularly in volatile market environments.

Research Questions

This study is guided by the following research questions:

- **RQ1:** Can LSTM-based return forecasts improve risk-adjusted portfolio performance (Sharpe/Sortino ratios) compared to classical MVO?
- **RQ2:** Are LSTM-enhanced portfolios more robust than MVO during high-volatility market regimes?
- **RQ3:** Does incorporating nonlinear temporal structure reduce downside risk relative to MVO?

Dataset

Historical market data will be retrieved using the yfinance Python library. The dataset consists of daily Adjusted Close prices for a diversified set of S&P 500 sector ETFs (e.g., XLK, XLF, XLE, XLY, XLV, XLP, XLI, XLB, XLRE). Adjusted Close prices ensure accurate return computation by incorporating dividends and stock splits. Volatility regimes will be identified using rolling standard deviation or VIX-based thresholds.

Methodology

Baseline: Markowitz Mean-Variance Optimization

The MVO framework (Markowitz, 1952) will serve as the baseline. Log returns and covariance matrices will be computed, and quadratic optimization will be used to derive efficient frontier weights. Static MVO portfolios will be evaluated under a rolling-window out-of-sample back-testing procedure.

Proposed Model: LSTM-Based Forecasting

The LSTM model (Hochreiter & Schmidhuber, 1997) will be trained on rolling windows of historical returns to forecast next-period returns. Forecasted returns will be used to construct portfolios through:

- proportional allocation based on predicted risk-adjusted returns, or
- forecast-augmented MVO optimization.

Hyperparameters such as sequence length, hidden units, dropout rate, and learning rate will be tuned using standard deep learning practices.

Backtesting and Evaluation

A walk-forward rolling-window backtesting framework will be implemented following best practices in financial evaluation. Performance will be assessed using:

- Sharpe ratio
- Sortino ratio
- Maximum Drawdown (MDD)
- Value-at-Risk (VaR) and Conditional VaR (CVaR)
- Portfolio turnover and transaction cost sensitivity
- Performance across volatility regimes

4. Project Timeline and Milestones

Milestone	Target Date
Data Collection & Preprocessing	3/15
MVO Baseline Implementation	3/20
Deep Learning Model Development	3/24
Mid-term Presentation (Results vs. Baseline)	3/31
Back testing & Comparative Analysis	4/20
Final Report Drafting & Visualization	5/05
Final Project Submission	5/12

Task Allocation

- Alexy: Data collection, preprocessing, and implementation of the MVO baseline.
- Sean: Development and tuning of the LSTM forecasting model.
- Tadiwanashe: Back testing engine, risk metric computation, comparative analysis, and visualization.

Conclusion

By comparing classical MVO with LSTM-based forecasting, this project aims to determine whether deep learning can meaningfully enhance portfolio optimization. The findings may provide valuable insights into the role of nonlinear modeling in modern quantitative finance.

References

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